

Helpful data: Atomic masses (g/mol): Mg=24, O=16, H=1, Co=59, Cl=35.5, C=12 g

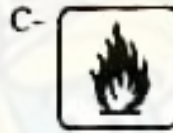
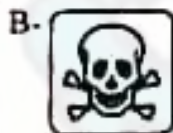
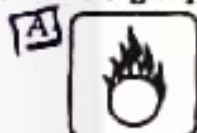
Choose the correct answer for the following questions and transfer it to the table in capital letters

Part 1 :	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Part2 :	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20

Q1: What is the CORRECT procedure for students to follow if a chemical is spilled?

- A- Turn off any gas or equipment and leave.
- ☒ B- Stand back and inform the teacher of the spill.
- C- Run madly out of the room.
- D- Sweep (مسح) it into your hand and throw it out.

Q2: The right pictogram that describe a flammable hazardous is:



Q3: Which one of the chemicals below will show clear solution with Hexane?

- A- NH_4Cl
- B- CaCO_3
- ☒ C- C_6H_6
- D- CH_3OH

Q4: A suitable equipment that used to measuring the volume of solid in density experiment is:

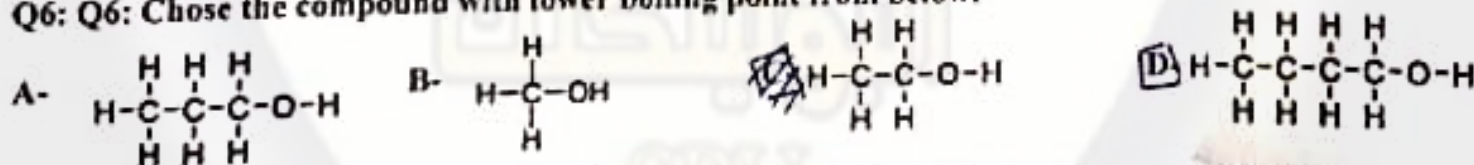
- A- Volumetric flask
- B- Dropper
- ☒ C- Balance
- D- Graduated cylinder

Q5: During Density experiment, the data in the table beside are obtained, calculate the final volume of water in the cylinder used in this measurement:

- ☒ A- 13.8 mL
- B- 14.3 mL
- C- 17.2 mL
- D- 17.7 mL

The density of solid	3.07 g/cm ³
Initial volume of water	11.9 ml
Mass of empty cylinder	25.3 g
Mass of cylinder + solid	41.5 g

Q6: Chose the compound with lower boiling point from below:



Q7: In an experiment, 15 g of red $\text{CoCl}_2 \cdot \text{XH}_2\text{O}$ is heated in a crucible. After heating produces a blue powder of CoCl_2 and a constant mass of 9.32 g is recorded. Calculate the water of crystallization in this salt?

- A- 6
- B- 8
- C- 4
- D- 7

Q8: Calculate the water percent ($H_2O\%$) in an alum if you have 7.52g of $NaAl(SO_4)_2 \cdot 12H_2O$.

(Mm of alum = 458 g/mol)

A- 26.05%

B- 46.89%

C- 39.08%

D- 47.16%

Q9: In empirical formula experiment, data beside were obtained:

The empirical formula of Magnesium oxide is:

A- MgO

B- Mg_2O

C- MgO_2

D- Mg_2O_2

Mass of crucible	23.4 g
Mass of crucible + Magnesium	26.2 g
Mass of crucible + Magnesium oxide	29.8 g

Q10: A compound has an empirical formula of $C_3H_4Cl_2$. Its molar mass is 113 g/mol. Find its molecular formula.

A- $C_3H_4Cl_2$

B- $C_6H_8Cl_4$

C- $C_4H_{12}Cl_4$

D- $C_6H_{12}Cl_2$

**Part 2: True or False Questions:

Q11: First aid kit is a suitable equipment that used to give medical treatments. *True*

Q12: After finishing the experiments, cleaning your working area must be all the time. *True*

Q13: The solubility of solid decrease when temperature increase. *False*

Q14: The measured density of a solid will be decrease if a part of the solid dissolved in the liquid used and the volume expand. *False*

Q15: Anhydrous is the name which given to a crystal salt that lost water of crystallization.

Q16: When magnesium reacts with air, it combines with another element besides oxygen and ammonia gas was liberated after adding water.

Q17: You can't heat ethanol to measured its boiling point on blue flame directly.

Q18: Capillary tube was used to measuring the boiling point temperature of unknown liquid.

Q19: Color change and glowing are indications of forming magnesium oxide in empirical formula.

Q20: The calculated (X) will be higher if you don't cool the crucible in desiccator.

Good Luck



10/22/2022 – Sunday***

Student name: [redacted] Instructor: [redacted] Section: [redacted] (11-2)

[Helpful information: $R = 0.0821 \text{ atm.L/mol.K}$; mwt: $\text{H}_2\text{O} = 18 \text{ g/mol}$; $760 \text{ mmHg} = 1 \text{ atm}$]

* Choose the correct answer for the following questions and transfer it to the table below:

Question	1	2	3	4	5	6	7	8	9	10
Answer	A ✓	B ✓	B ✓	D ✓	C ✓	A ✗	C ✗	B ✓	C ✗	B ✓

Question	11	12	13	14	15	16	17	18	19	20
Answer	B ✓	C ✓	C ✓	C ✓	A ✓	A ✓	A ✓	D ✓	C ✓	B ✓

1- Which of the following statements is correct?

- A. When dealing with alcohols; heat with indirect flame by using water bath.
- B. Return used chemicals to the stock bottles.
- C. Hold your face directly over the container when you smell a chemical substance.
- D. It's allowed to withdraw samples form the stock solution bottles with your own pipet.

2-The following set of data were obtained by student during measurement density of solid material, Calculate Mass of the solid: -----:

- A. 0.30 kg
- B. 0.33 kg
- C. 3.00 g
- D. 13.3 g

$$\rho = \frac{m}{V}$$

density of solid = 5 g/cm^3

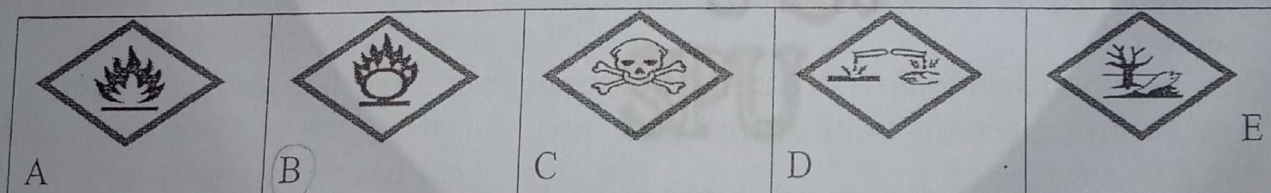
Initial volume of water in cylinder: 18.5 ml

Final volume of water in cylinder: 85.0 ml

$$332.5 \times 10^{-3}$$

$$5 = 66.5$$

3-The pictogram that correctly indicates Oxidizing chemical hazard is:



4- Given the following set of data:

- Mass of empty crucible = 15.0g.	- Mass of crucible + alum = 16.2 0g.
- Mass of crucible + anhydrous salt = 15.8g	- Molar mass of anhydrous salt = 258.0g/mol.

Calculate water of hydration (X) in the alum.

- A. 13 B. 14 C. 8 D. 7

5- If (X= 10) and you have the following data calculate mass of hydrates salt

- A. 0.73 g
B. 1.75 g
C. 1.22 g
D. 0.68 g

Mass of empty crucible	5.21 g
Mass of crucible and anhydrous	6.23 g
Molar mass of anhydrous	150 g/mol
Molar mass of H ₂ O	18 g/mol

6- Calculate the temperature (in° C) of 2.0g of a vapor of a volatile Liquid (condensed vapor) (molar mass 90.0g/mol) which occupies a volume of 700.0 ml at a pressure of 720 mmHg:

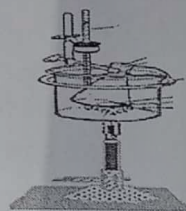
- A. 363 °C B. 225°C C. 519 C° D. 90 °C

7- Regarding the boiling point which one is incorrect?

- A. It is temperature at which vapor pressure of liquid exactly equals external pressure
B. As external pressure increases the boiling point increase
C. As The molar mass of liquid molecule increases the boiling point decrease
D. It is used to check purity of compound

8- In which experiment dose this apparatus used:

- A. Boiling point identification B. Density identification
C. Molecular weight of volatile liquid D. Water of crystallization



9- The vapor of an unknown volatile liquid occupies a flask at 100.0°C and 1 atm, if the molar mass of volatile liquid equal 40.0 g/mol. what is the density of this volatile liquid?

- A. 4.9 g/L B. 1.3 g/L C. 2.3 g/L D. 5.9 g/L

10- The molecular weight can be determined by dumas method for which of the following liquids

- A. Chlorobenzene (B.p 132)
B. Ethanol (B.p 78)
C. Toluene (B.p 110)
D. Heptane (B.p 102)

11- The boiling point of a liquid is dependent on pressure, which one of the following is correct

- A. Yes, negative relationship with external pressure
B. Yes, Positive relationship with external pressure
C. No, does not effected
D. No, negative relationship with vapor pressure

12- Which of the following statement is incorrect?

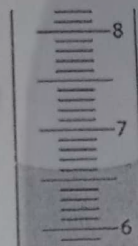
- ✓ A. $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ lower boiling point than $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
- ✓ B. $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ higher boiling point than $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
- ✓ C. CH_3OH lower boiling point than CH_3CH_3
- ✓ D. $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ higher boiling point than $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$

13- Regarding the accuracy of the balances which one is correct?

- A. Top loading balance (with accuracy ± 0.0001)
- B. Analytical balance (with accuracy ± 0.001)
- ✓ C. Analytical balance more accurate than Top loading balance
- D. Analytical balance (with accuracy ± 0.01)

14- What is the volume inside the cylinder?

- A. 6.5 ml
- B. 66 ml
- ✓ C. 6.6 ml
- D. 65 ml



15- One of the following salt is insoluble in H_2O (polar solvent)?

- ✓ A. AgCl
- B. KI
- C. K_2SO_4
- D. NaCl

16- In a solution, what term is given to the solid that has dissolved?

- ✓ A. Solute
- B. Solvent
- C. Reagent
- D. Soluble

17- Expression for molar mass determination of volatile liquid (Hint's; d: density).

- ✓ A. dRT/P
- B. mRT/dP
- C. $mVRT/P$
- D. RT/mPV

$$PV = TRn$$

$$PV = TR \frac{m}{M_w}$$

$$M_w = \frac{TRd}{PV}$$

18- During molecular weight of volatile liquid experiment. In the end of experiment, we weight.....and the volume of vapor take by using.....? (Hint's; cover: Al foil and rubber)

- A. mass of empty flask and cover - cylinder
- B. mass of empty flask and cover - flask
- C. mass of flask, cover and vapor - flask
- ✓ D. mass of flask, cover and vapor - cylinder

صعود السائل

19- During the hydrated salt Experiment how mass of hydrate salt effected if we put sample in desiccator?

- A. Increase
- B. Decreases
- C. No effected

20- For most solid, what happens to the Density as the temperature increases?

- A. Increases
- B. Decreases
- C. No effected

$$PV = R_n T$$

Good luck

$$\frac{PV}{V} = \frac{RT \cdot \frac{m}{mw}}{V}$$

$$P = \frac{RT \cdot d}{mw}$$

$$\frac{P \times mw}{R d \downarrow} = T \uparrow$$



Student name :	Instructor :	Section:
----------------	--------------	----------

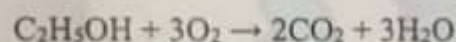
Helpful data: 1 atm = 760 mmHg = 101.3 Kpa; $R = 0.0821 \text{ L.atm. Mol}^{-1}.\text{K}^{-1}$

Atomic masses (g/mol): Mg=24, O=16, C=12, H=1, Na=23, Cl=35.5, S=32,

Choose the correct answer for the following questions and transfer your answer to the table in capital letters and commit to significant figures

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13
Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22	Q23	Q24	Q25	

Q1- Calculate ΔH for the following reaction



Use the following data in your calculation

$\text{C}_2\text{H}_5\text{OH} \rightarrow 2\text{C} + 3\text{H}_2 + \frac{1}{2}\text{O}_2$	$\Delta H = 228 \text{ kJ/mol}$
$\text{C} + \text{O}_2 \rightarrow \text{CO}_2$	$\Delta H = -394 \text{ kJ/mol}$
$\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O}$	$\Delta H = -286 \text{ kJ/mol}$

- A) -452 KJ/mol B) -1418 KJ/mol C) -298 KJ/mol D) -1076 KJ/mol

Q2- When 50.0 g of 0.200 M $\text{NaCl}(aq)$ at 24.1°C is added to 100.0 g of 0.100 M $\text{AgNO}_3(aq)$ at 24.1°C in a calorimeter, the temperature change to 25.2°C as $\text{AgCl}(s)$ forms. Assuming the specific heat of the solution and products is $4.20 \text{ J/g}^\circ\text{C}$, calculate the approximate amount of heat of the solution in joules.

- A) 693 J B) - 462 J C) -693 J D) 462 J

Q3- When 1.5 g of fructose, $\text{C}_6\text{H}_{12}\text{O}_6$, is burned in oxygen in a bomb calorimeter, the temperature of the calorimeter is decreases about 1.58°C . If the heat capacity of the calorimeter and its contents is $9.90 \text{ kJ/}^\circ\text{C}$, what is q_{rxn} for this combustion? and is this reaction endothermic or exothermic?

- A) -15.64 kJ, exothermic reaction B) 15.64 kJ, endothermic reaction
C) -12.34 kJ, exothermic reaction D) 12.34 kJ, endothermic reaction

Q4- Which one of the following is correct about your thermodynamic experiment?

- A) A bomb calorimeter was used to measuring the heat of reaction.
- B) The heat gained by calorimeter equal zero.
- C) A cup calorimeter was used to measuring the temperature change of solution.
- D) The atmospheric pressure was affected your experiment.

Q5- A 10.0 mL bleach sample was diluted to 100 mL in a volumetric flask. A 25.0 mL of this solution was titrated with 0.30 M $\text{Na}_2\text{S}_2\text{O}_3$ to reach the endpoint. If $\text{Cl}_2\%$ = 2.35% in original sample, Calculate the final volume of $\text{Na}_2\text{S}_2\text{O}_3$ if the initial volume was 7.5mL (Molar mass NaClO = 74.5 / Cl_2 =71 g/mol and the density of bleach solution is 1.08g/mL).

- A) 11.9 mL
- B) 13.5 mL
- C) 19.4 mL
- D) 9.2 mL

Q6- The active ingredient in bleach is:

- A) $\text{Na}_2\text{S}_2\text{O}_3$
- B) KI
- C) CH_3COOH
- D) NaOCl

Q7- The color of bleach solution was change during titration from _____ to _____ during adding indicator.

- A) Brown – yellow
- B) Brown – Color less
- C) Yellow – Dark blue
- D) Dark blue – Color less

Q8- In bleach analysis experiment, if you add 40 mL water instead of 25 mL, then the volume of $\text{Na}_2\text{S}_2\text{O}_3$ used to reach the end point will be:

- A) Increased
- B) Decreased
- C) Be the same

Q9-The indicator used during vinegar analysis has a----- color in the acidic medium:

- A) Colorless
- B) Pink
- C) Brown
- D) Blue

Q10- A 30 mL (5.2 M) vinegar sample was titrated with (3.75 M) of NaOH solution, then the volume of titrant needs to reach the endpoint equal:

- A) 41.6 mL
- B) 21.6×10^{-3} mL
- C) 21.6 mL
- D) 41.6×10^{-3} mL

Q11- A 20 mL vinegar sample ($d = 1.0$ g/mL, molar mass of acetic acid= 60.0 g/mol) has been titrated with a 1.2 M NaOH solution, If the initial reading of the burette was 4.2 mL, and the final burette reading was 17.3mL, then the % CH_3COOH in vinegar solution equals:

- A) 4.72%
- B) 47.2%
- C) 4.01%
- D) 40.1%

Q12- A suitable equipment used to determine the volume of sodium hydroxide in acid-base titration is:

- A) Dropper
- B) Burette
- C) Cylinder
- D) Receiving flask

Q13- 48g of silver nitrate (AgNO_3) ($M_m = 170\text{g/mol}$) was react with 32g of sodium chloride (NaCl) ($M_m = 58.4\text{g/mol}$) according to the equation below, if the mass of solid collect = 17.8g then the % of the precipitate is ($M_m \text{NaNO}_3 = 85$ / $\text{AgCl} = 143.3 \text{ g/mol}$):



- A) 74.8 % B) 22.6% C) 44.4 % D) 56.5%

Q14- The Solid material formed in the reaction $\text{Na}_2\text{CO}_3 + \text{CaCl}_2 \longrightarrow \text{CaCO}_3(s) + 2\text{NaCl}$ was:

- A) Filtered, washed with cold water, and then allowed to dry.
 B) Washed with cold water t, Filtered, and then allowed to dry.
 C) Allowed to dry, washed with cold water, and then Filtered
 D) All of the above

Q15- If you forget drying the sample of precipitate and taking its final weight as your exp. procedure, then the calculated yield% will be:

- A) Increase B) Decrease C) Not affected

Q16- The mass of carbon dioxide (CO_2) gas occupying a 5.604 L container at 28.2°C and 92011.3 Pascal equals?? (Assume it behaves as an ideal gas)

- A) 10.97 g B) 1.097 g C) 90.57 g D) 9.057 g

Q17- During the molecular weight of volatile liquid experiment. At the end of experiment we record the.....and then the volume of gas measured by using.....

- A) Flask +condensed vapor /conical flask B) Flask + condensed vapor / cylinder
 C) empty flask / cylinder D) empty flask/ Volumetric flask

Q18- Which one of these compounds you can applied dumas method to calculate its molecular weight?

- A) Acetone (B.P= 56°C^*) B) Toluene (B.P= 121°C^*)
 C) Chlorobenzene (B.P= 131°C^*) D) Hexanal (B.P= 129°C^*)

Q19- Compound X contains only C,H,O and S. A 5.00 g sample undergoes combustion to give 4.83 g of CO_2 and 1.48 g of H_2O and a certain amount of H_2SO_4 which titrated with 0.109 mol of 1.0M NaOH. What is X? (Assume each mole of H_2SO_4 equivalenced with 2 mole NaOH)

- A) $\text{C}_2\text{H}_3\text{SO}_2$ B) $\text{C}_3\text{H}_3\text{SO}_2$ C) $\text{C}_2\text{H}_3\text{SO}_2$ D) $\text{C}_2\text{H}_3\text{SO}_3$

Q20- In empirical formula of magnesium oxide exp., if some oxide was splashed out the crucible before final weight, then the calculated ratio of Mg/O will be:

- A) Increase B) Decrease C) Be the same

Q21- For the hydrated salt: $\text{MgSO}_4 \cdot x \text{H}_2\text{O}$, if the mass of hydrated salt is 2.0 g and % H_2O = 47.3%, then the value of (x) is:

- A) 15 B) 5 C) 10 D) 6

Q22- A student knows the mass of solid object. What other variable does the student need to know to calculate the object's density?

- A) The volume of object and it measured by ruler.
B) The volume of object and it measured by volumetric flask.
C) The volume of object and it measured by graduated cylinder.
D) The volume of object and it measured by Erlenmeyer flask.

Q23- At which of the following external pressures will benzene has its lowest boiling point ?

- A) 760 torr B) 755 torr C) 785 torr D) 792 torr

Q24- KNO_3 is ----- in water (H_2O)

- A) Soluble B) Insoluble C) partial soluble

Q25- Which of the following pairs are the least precise volume measurement equipment of liquids?

- A) Beaker and burette
B) beaker and graduated cylinder
C) Beaker and conical flask
D) Pipet and burette

Good Luck
in
your exams